V8 | ALLEN NEUROPIXELS | EXPLORATORY REAL-DATA BENCHMARK

Information processing in real neural recordings Classical and quantum-like models

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Post-flash activity carries decodable stimulus information.

Classical models match or outperform quantum-like models in this pilot.

01 / Real recordings and held-out tests

Allen Visual Coding Neuropixels: 3 mice,
450 black/white flashes, 131,080 exact spikes.
LGd (thalamus), V1 and VISl (visual cortex).
Primary comparison: 2 mice with all 3 regions;
48 units each, 30 test flashes per mouse.
A third mouse lacks V1 flash-block coverage.
20 ms bins; chronological 90/30/30 trial split.
All fitting and model choices use train/validation
trials. Readout: 350–600 ms after onset
(100–350 ms after the flash ended).

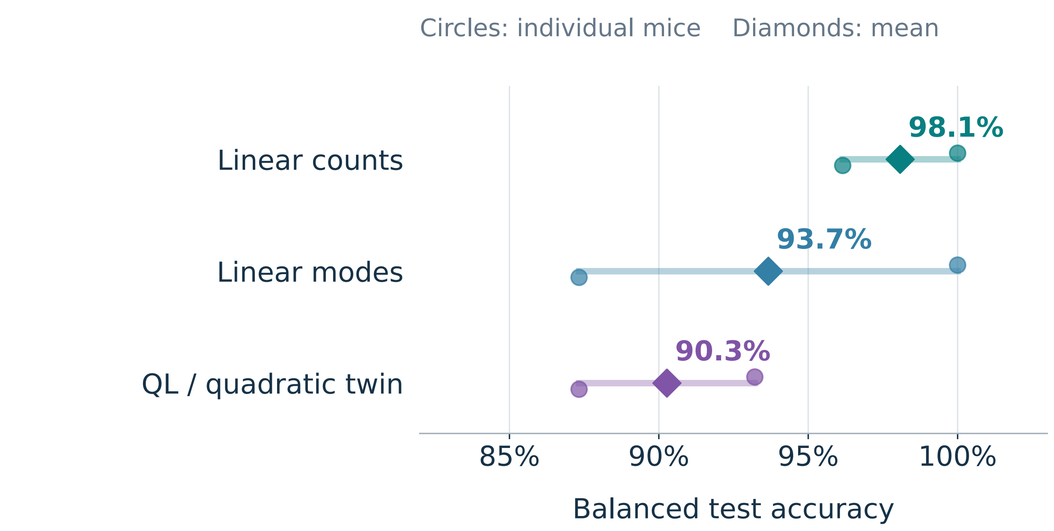
02 / A testable quantum-like representation

Scholes-inspired collective modes encode
present and past activity in complex coordinates.
No microscopic quantum phase was measured.

$$R' = T\,R\,T^{\dagger} \qquad \text{with } R = q\,q^{\dagger}$$

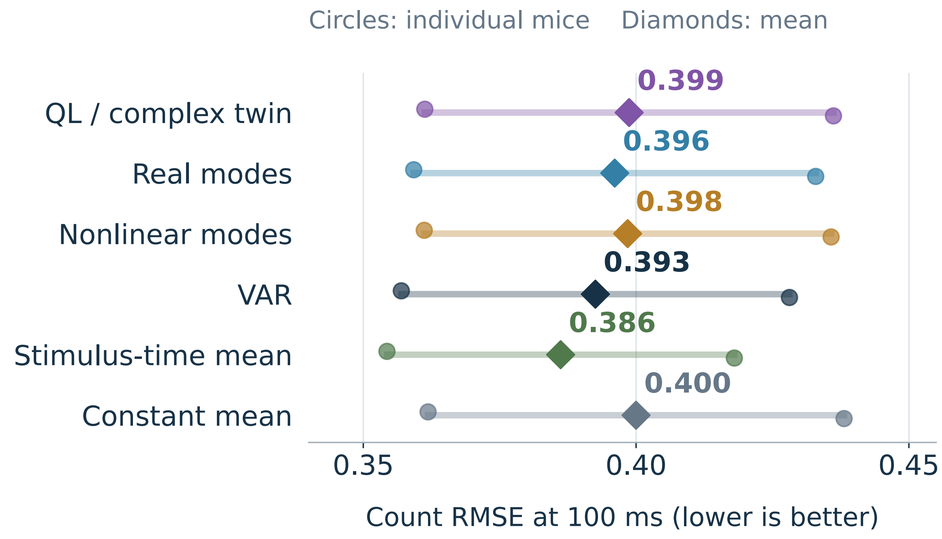
The anchored state q = [a, 1] preserves amplitude.
Density predictions equal a classical complex twin
by construction; the density readout also has
an equivalent classical quadratic feature map.
Controls: real and nonlinear modes, VAR,
stimulus-time mean (PSTH), and constant mean.

03 / Post-flash stimulus information



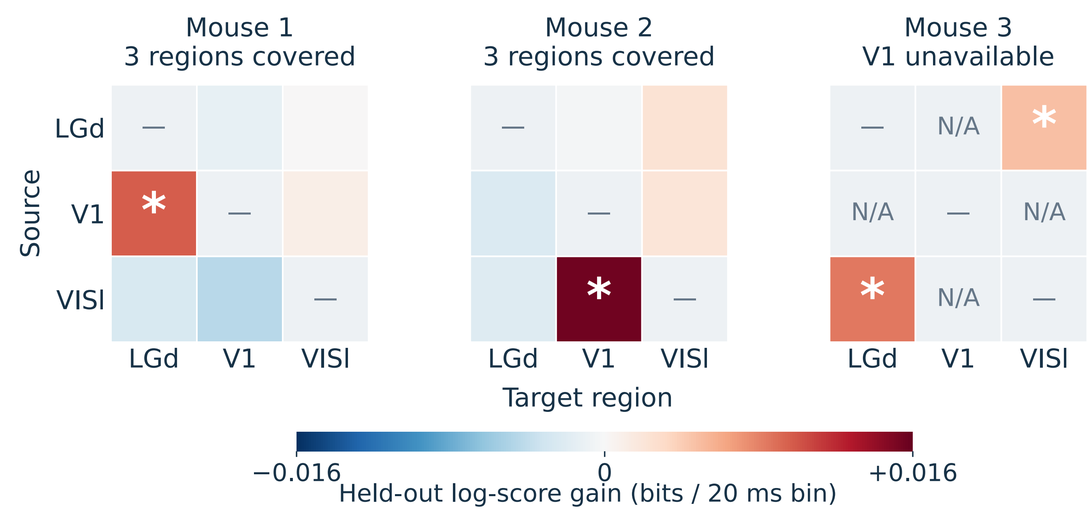
Overall polarity accuracy: 98.3% for linear counts,
93.3% for linear modes, 90.0% for density/twin.
OFF/rebound responses can explain this signal;
this was not a controlled working-memory task.

04 / Prediction at 100 ms



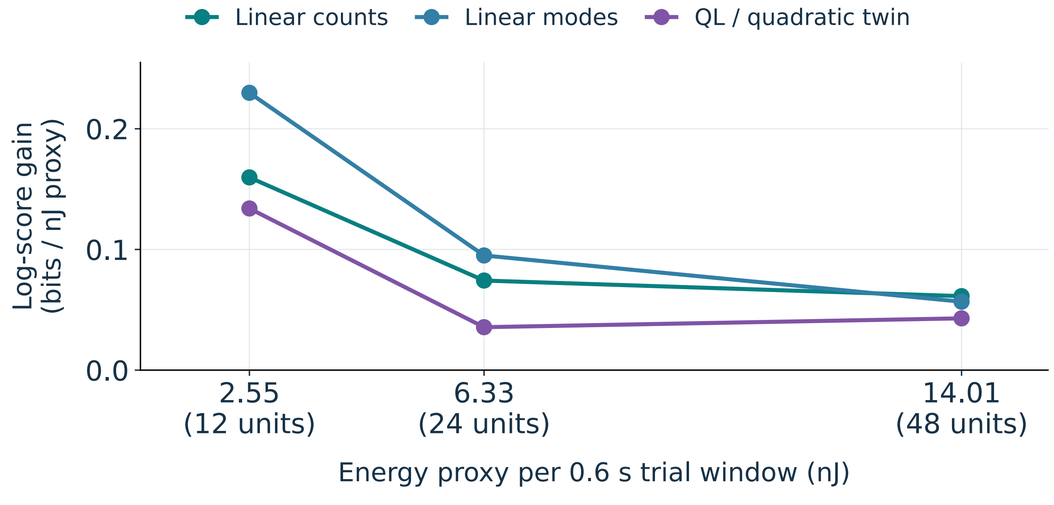
Classical VAR and stimulus-time mean have lower
mean RMSE. Density/complex uses 314 fitted
coefficients versus 11,568 for VAR, but a constant
mean is already close (0.400; 48 coefficients).

05 / Directed predictive dependence



Conditional Gaussian log-score proxy; past 20–60 ms.
* q < 0.05: 4/14 links; 999 trial shuffles, BH correction.
Evoked-response and running controls; patterns differ.
Mouse 3 lacks V1 conditioning; N/A links are excluded.

06 / Information per energy proxy



Shared recorded-unit ATP cost; no measured metabolism.
48 units: 0.061 bits/nJ (counts) vs 0.043 (density/twin).
Sampling is not a brain saving. Density dynamics store
34 real scalars vs 6 for their classical amplitude twin.

Interpretation

A reproducible pilot for comparing collective-mode representations.
Directed dependence is not causal routing; ATP proxies do not establish
metabolic savings. Larger independent cohorts are the next test.

Sources: Siegle et al., Nature (2021); Scholes, Proc. R. Soc. A (2024).
Schreiber (2000); Barnett et al. (2009); Howarth et al. (2012).